

EXAM 7 – SPRING 2014

25. (2.5 points)

a. (1 point)

Define “bridging model” in the context of the loss ratio planning process and describe an operational problem created when a bridging model is used.

b. (1.5 points)

Identify two aspects of effective cycle management and describe how focusing on these aspects can help mitigate operational risks through underwriting cycles.

CONTINUED ON NEXT PAGE

SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 25	
TOTAL POINT VALUE: 2.5	LEARNING OBJECTIVE: C7: Describe operational risk and demonstrate possible mitigation and quantification methodology
SAMPLE ANSWERS	
Part a: 1 point	
<i>Sample Answer 1</i> A bridging model uses mature prior year loss ratios, combined with loss cost and rate level changes, to project loss ratios for more recent years. This introduces significant autocorrelation between loss ratios in the planning process and older loss ratios... <i>Sample Answer 2</i> Bridging Model: loss ratios for older more mature years are used as expected losses for recent years, adjusting for inflation and trends. For recent years, a BF method is used to weigh actual experience and the expected loss using the older year ELR. Optimistic older LR's lead to optimistic plan LR's. An operational prob exists for a long tailed line, where deterioration in older year loss ratios gets carried forward to recent years, causing a reserve conflagration.	
Part b: 1.5 points	
<i>Sample Answer 1</i> 1. Intellectual property: Includes investment in talent, systems and databases, and comprises majority of franchise value. Insurers should invest in intellectual property throughout the u/w cycle through investment in talent, systems, and databases. This mitigates risk of systems/databases becoming obsolete and talent leaving especially during a soft market. 2. Underwriter incentives – Underwriter incentives should be flexible and aligned to the goals of the portfolio. If market prices become inadequate during a soft market, underwriters should not be penalized for ceasing to write new business if this is in the portfolio's best interest. <i>Sample Answer 2</i> 1. Firm's franchise values are mainly from intellectual property including experts in the underwriting, actuarial, etc., proprietary systems for policyholder information, forecasting systems, reputations, market relationship. To mitigate operational risk through cycle, firm needs to protect this intellectual property through retaining top talent, invest in systems, maintain presence in the core market so when the cycle goes back to favorable environment, the firm still has these assets to help create more value. 2. Owner's' education. Under bad market, premium won't increase as much while expense is still growing, these caused certain ratios such as expense ratios increase. Company's management and shareholders should have the perspective that this is due to the cycle so that they won't make wrong moves such as cutting premium to gain market share which	

SAMPLE ANSWERS AND EXAMINER'S REPORT

will bring the firm to the vicious cycle that not able to retain future earnings. So a good understanding from owners will help mitigate the operational risks through the cycle.
EXAMINER'S REPORT
Part a
Candidates were expected to know the definition of a bridging model. Many candidates recognized that a bridging model was an approach to estimate plan from historical loss ratios but did not give any details of the adjustments necessary in this process (for trend, rate level, etc.). Some candidates incorrectly described a bridging model as a way to estimate reserves from the plan loss ratio, or else they related reserving to pricing. Candidates received credit for the operational risk portion of the question if they related the risk to use of the bridging model. Generally, full credit answers identified the correlation between years that result from using a bridging model.
Part b
Candidates were expected to understand underwriting cycle management and know what companies should do to operate effectively through the cycle. Candidates generally scored well on this section. Some candidates used two different examples of intellectual property. This was considered a duplicate answer and did not receive full credit. Some candidates identified an aspect of intellectual property but did not describe how it could help mitigate risk.